Germania Nanoparticles and Nanocrystals at Room Temperature in Water and Aqueous Lysine Sols

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Facile synthesis of nanometer-sized germania crystals and amorphous germania nanoparticles (ca. 1 nm) is investigated through hydrolysis of germanium tetraethoxide and subsequent condensation of germania in both pure water and aqueous lysine solutions. Germanium tetraethoxide rapidly hydrolyzes in pure water, leading to solvated germanate species at lower germania concentrations and the onset of nanometer-sized germania crystals at room temperature with increasing germania content. In the presence of the basic amino acid L-lysine, amorphous germania nanoparticles (ca. 1 nm) spontaneously form with increasing germania content and coexist with nanometer-sized germania crystals at higher germania concentrations. Lysine and germania concentration both influence crystallite size and morphology (i.e., polyhedral, cubic). The facile, room-temperature crystallization of germania in the presence and absence of lysine is striking. The fact that the crystal morphology shows no signs of nanoparticle aggregative assembly, as has been observed in the formation of other oxide crystals, suggests that crystal growth takes place by addition of dissolved species rather than nanoparticles, and could have implications for other oxide systems.

I. Introduction

Studied extensively decades ago (e.g., refs 1−3), the oxide of germanium has experienced a recent revival, often in conjuction with its silica analogue, owing to its enhancement of material properties ranging from reactivity[4] to optics.[5,6] Motivated by the importance of silica nanoparticles in zeolite crystallization[7] and the improved catalytic properties of zeolite frameworks that incorporate germanium, Rimer et al.[8] recently explored the chemistry of germania in highly alkaline mixtures containing various common zeolite structure directing agents. Their work highlighted similarities between the phase behavior of highly basic germania and silica sols, commonalities that have further impacted the area of biomineralization. Specifically, the copolymerization of silica and germania in biogenic systems has been observed,[9] evidence exists for the importance of germanium in biosilicification,[10] and it is recognized that similar biomineralization mechanisms may govern synthesis of complex germania structures as well.[11−13]

Recently, we showed that hydrolysis of tetraethylorthosilicate (TEOS) in aqueous lysine solutions[14] leads to the formation of fairly monodisperse silica nanoparticles in the range 5−25 nm, which can be ordered into nanoparticle arrays.[15,16] Interested in determining how the aqueous chemistry of germania compares to that of silica, and spurred by an interest in exploring the influence of biomolecules on germania crystallization and nanoparticle formation, we report here on the condensation of germania in aqueous mixtures in the absence and presence of the basic amino acid L-lysine.

II. Experimental Section

Germanium (IV) tetraethoxide (GTE, 99.95%, Aldrich), purified water (Millipore Elix, 10 MΩ cm), and L-lysine (Sigma) were combined to give molar compositions g GeO_2/x lysine/9500 H_2O/4g ethanol where $0 \leq g \leq 60$ and $x = 0$, 5.8. Complete hydrolysis of GTE was carried out under vigorous stirring at room temperature for approximately 20 h. Small- and wide-angle X-ray scattering (SWAXS) were employed to simultaneously characterize the resulting sols with a SAXSess instrument (Anton Paar GmbH) employing Cu K_α radiation. Data analysis accounted for water as the solvent, and the indirect Fourier transform technique of Glatter[17,18] was applied to extract pair distance distribution functions (PDDF) and estimate particle size. Cryogenic transmission electron microscopy (cryo-TEM) was employed for further in situ imaging of the sols. Samples were prepared according to the method described previously.[19,20] Images were collected with a JEOL 1210 TEM operating at 120 kV equipped with a liquid nitrogen cooled specimen stage. Mass spectra were collected with a Bruker BioTOF II electrospray ionization instrument with a capillary exit voltage of −110 V, spray voltage of 4000 V, desolvation temperature of 200 °C, and continuous sample flow rate of 150 μL/h. SEM images of dried droplets of the sols (i.e., without dilution or purification) coated with ∼30−50 Å of platinum were collected on a JEOL 6500 instrument. Crystalline precipitates

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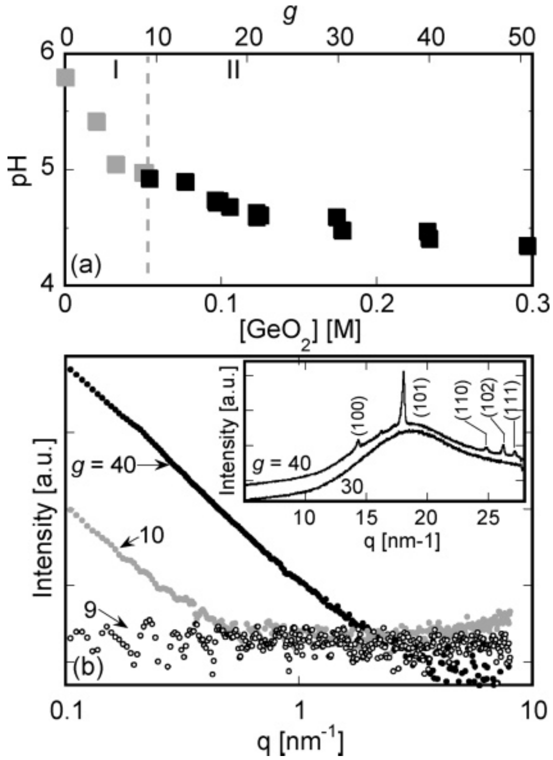

Figure 1. (a) pH as a function of germania composition for water−germania sols. Symbol shading represents correspondence to two distinct regimes (I and II): (I) sols with germanate species (closed gray symbols) and (II) crystalline germania particles (closed black squares). (b) Representative SAXS data showing the evolution of the low-*q* scattering with increasing germania concentration for molar composition *g* GeO_2/9500 H_2O/4*g* ethanol, with $g = 9$, 10, and 40. The inset shows SWAXS data for two sols of region II, with labeled reflections corresponding to the crystalline hexagonal phase of germania.

were isolated by centrifuging the sols, withdrawing the supernatant, and drying the samples at room temperature under vacuum. Powder diffraction patterns of dried precipitates were measured on a Bruker-AXS (Siemens) D5005 diffractometer with Cu K_α radiation.

III. Results and Discussion

Hydrolysis of germanium (IV) tetraethoxide (GTE) was found to occur rapidly in water in the absence of any other hydrolytic agents. As shown in Figure 1a, the pH of the resulting aqueous germania mixtures decreases as a function of increasing germania content. The linear decrease in pH at lower germania concentrations resembles that of TEOS hydrolysis,[14] and can be similarly explained through the following series of reactions:[8]

$$Ge(OC_2H_5)_4 + 4H_2O \rightarrow Ge(OH)_4 + 4C_2H_5OH \quad (1)$$

$$Ge(OH)_4 + OH^- \rightarrow Ge(OH)_3O^- + H_2O \quad (2)$$

$$Ge(OH)_3O^- + OH^- \rightarrow GeO_2(OH)_2^{\ 2-} + H_2O \quad (3)$$

The decreasing pH is also consistent with previous studies by Kawai et al.[21] that confirmed that the rapid hydrolysis of GTE forms soluble species, the majority of which are in the form of deprotonated germanic acid. A transition in the pH sensitivity of the aqueous germania solutions studied here occurs at approximately 0.06 M germania concentration (Figure 1a).

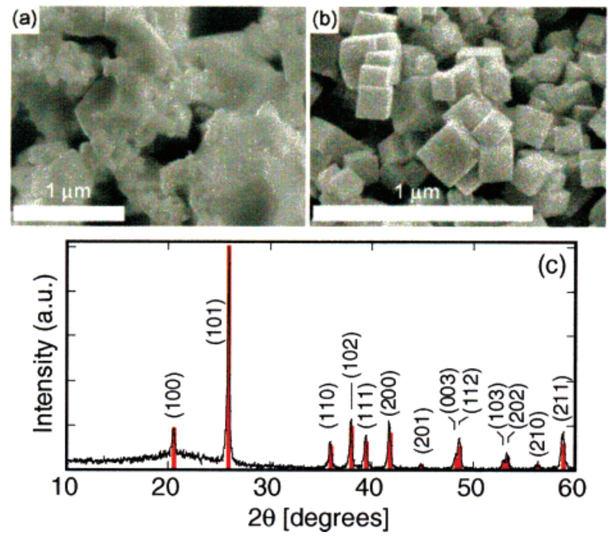

Figure 2. SEM images of dried precipitate from sols with molar composition *g* GeO_2/9500 H_2O/4*g* ethanol where (a) $g = 20$ and (b) $g = 30$. Panel (c) shows the powder XRD of precipitate from the $g = 40$ sol compared to the theoretical diffraction pattern (red) of the crystalline hexagonal phase of GeO_2.

Although subtle, this transition is accompanied by a dramatic change from clear solutions to turbid ones of increasing opacity, beginning approximately 1 h after the reaction is initiated. SAXS analysis of the water−germania mixtures studied here results in the identification of two distinct regimes by increasing germania concentration, which are indicated as regions I and II in Figure 1a. Namely, SAXS (Figure 1b) reveals an onset of a population of large particles at the transition point, denoted by an increase in low-*q* scattering for sols of region II. The absence of any scattering from samples of region I (e.g., $g = 9$) is consistent with the presence of only solvated small germanate species undetectable by SAXS.

For the germania−water system, SWAXS analysis of the sols (inset, Figure 1b) reveals the development of Bragg reflections at large *q* values for sols of the highest germania content explored here (e.g., $g = 40$). These reflections share identical positions to the low-angle peaks of the complete diffraction pattern obtained via powder XRD of the dried sols (Figure 2c). This confirms the remarkable room-temperature, facile formation of crystalline germania particles (i.e., hexagonal phase) at high germania concentrations, and identifies the source of the onset of increased low-*q* scattering (Figure 1b) for those samples.

While SEM images of all dried sols from region II (e.g., $g =$ 20 and 30, Figure 2a,b; Figure S1, Supporting Information) show the presence of nanometer-sized crystals, Bragg reflections are only observed via SWAXS for sols of germania content exceeding $g = 40$ (inset, Figure 1b). However, this absence of Bragg reflections is attributed to the lower crystal concentration in the former samples, as the presence of crystals has been confirmed by in situ cryo-TEM imaging for sols assigned to region II. This in situ technique, employed to discount concentration-induced crystallization upon evaporatative drying, is described further later.

As shown in Figure 2a,b, crystal size and morphology are both sensitive to germania content. At relatively low germania concentrations ($g = 20$; Figure 2a), few large faceted crystals of polyhedral morpholgy are observed, along with small ill-defined particles. Increasing germania content ($g = 30$) results in particles of decreasing size that bear a more cubic morphology (Figure 2b). SAXS (Figure 1b) provides in situ evidence of this decreasing particle size as a shift in scattering to higher *q* values

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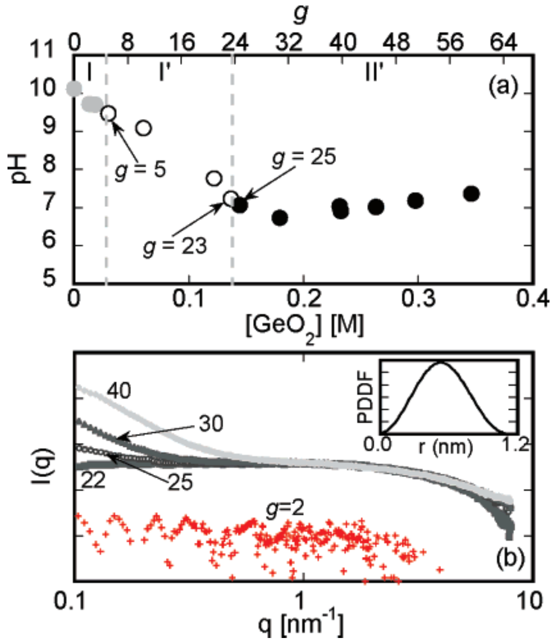

Figure 3. Evolution of pH (a) and representative SAXS patterns (b) as a function of germania content for sols having molar composition g GeO_2/5.8 lysine/9500 H_2O/4g ethanol. Symbols in (a) have been shaded to reflect correspondence to three distinct regimes (I, I′, and II′): (I) no particles detected (closed gray circles), (I′) nanoparticles (ca. 1 nm) (open circles), and (II′) coexistence of nanoparticles and crystalline particles. The inset in (b) shows a PDDF extracted from the scattering curve for the g = 20 sol.

with increasing germania content. It is conceivable that the number of nuclei increases with increasing germania content, resulting in the observed decrease in crystal size.

Motivated by the work of Rimer et al.[8] and our recent work exploring hydrolysis of TEOS under more neutral conditions imparted by the amino acid L-lysine, we focus for the remainder of this work on hydrolysis of GTE in the presence of lysine. For convenience, the resulting sols are referred to hereafter as "Lys-Ge." Figure 3a shows the sol pH as a function of germania content in Lys-Ge sols. The pH decreases with increasing germania concentration until a point of transition, defined by a dramatic change in pH sensitivity. These results are qualitatively similar to the behavior in the aqeous solutions described above and the results reported by Rimer et al.[8] for hydrolysis of GTE in the presence of TPA, TMA, and Na cations.

In contrast to GTE hydrolysis in pure water, however, SAXS analysis of the corresponding sols containing lysine (Figure 3b) reveals two distinct subregimes of the linear, low-germania portion of the titration (i.e., preceding the pH transition). We have labeled these as regions I and I′ in Figure 3a according to the form of germania in the sols. Specifically, sols in region I are composed only of dissolved germanate species, which are not detectable by SAXS (e.g., g = 2, Figure 3b). While the exact speciation may differ as a result of lysine, this initial formation of solvated germanate species at low germania concentrations is generallly consistent with the low-germania sols of the lysine-free system described previously. We address the identity of these species below using electrospray ionization mass spectrometry (ESI-MS).

With increasing germania content, nanoparticles are detected by SAXS in the Lys-Ge sols (e.g., g = 22, Figure 3b). The sols in this region (I′, Figure 3a) contain a population of nanoparticles of approximately 1 nm in size as estimated from the PDDF (inset, Figure 3b) extracted from the corresponding SAXS patterns. Nanoparticle size appears to be independent of the germania concentration in the range studied. While these

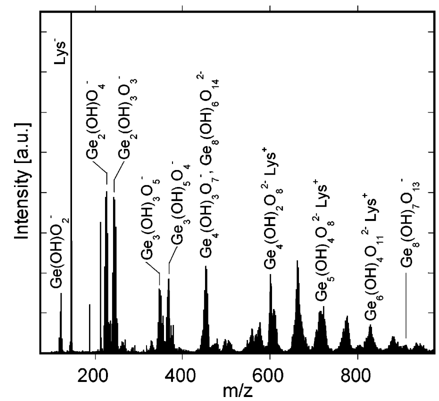

Figure 4. Mass spectrum indicating possible germanate species from a sol with molar composition 20 GeO_2/5.8 lysine/9500 H_2O/80 ethanol.

exceedingly small entities could not be resolved in cryo-TEM images, the size is consistent with the results for the germania-(TPA, TMA, and Na) systems.[8] In that work, the small nanoparticles were proposed to be double four-membered rings, $Ge_8O_{12}(OH)_8$.

We have employed ESI-MS to elucidate specific germanate speciation through detection of the negatively charged germanate species. Figure 4 shows the ESI mass spectra for a representative sol having germania composition g = 20. While the presence of the octamer (i.e., $Ge_8O_{14}(OH)_6{}^{2-}$) is detected, many other dominant species are observed, including germania–lysine complexes, suggesting that germania nanoparticles may assume a range of morphologies beyond cubic in the Lys-Ge sols.

This onset of detectable nanoparticles in Lys-Ge sols at germania concentrations ($g \geq 5$, open circles of region I′ in Figure 3a) well below that at which the pH transition occurs is in striking contrast to studies of germania in highly alkaline systems.[8] In the latter systems, nanoparticles were detected only at germania concentrations above the point of pH transition, interpreted as a "critical aggregation concentration" at which germania self-assembles to form nanoparticles. It is interesting to note, however, that the germania concentration at which nanoparticles first appear in the Lys-Ge sols ([GeO_2] = 0.03 M; g = 5) is similar to the reported germania solubility limit in water (~0.043 M).[22,23] Better correlation to the solubility limit may even exist, since the dependence of germania solubility on pH, particle size, and structure is unknown. A similar onset of nanoparticles upon exceeding the oxide solubility limit has been observed previously in the silica–TPA[7] and silica-lysine[14] systems. Possible reasons for the continued reduction of pH even after the onset of nanoparticle formation include the following: further deprotonation of germanic acid (reaction 3), and continued deprotonation of the formed nanoparticles.

After the transition point ([GeO_2] ≈ 0.14 M; g = 25) is reached, Lys-Ge sols of higher germania concentration (solid black symbols in region II′, Figure 3a) become turbid after only 1 h of reaction. As observed for lysine-free germania–water sols with high germania concentration, the steeper slope of the scattering curves at low q for these sols (Figure 3b) suggests the presence of a second population of larger particles. In the case of the Lys-Ge sols, this enhanced low-q scattering is accompanied by slightly increasing sol pH with increasing germania concentration. Increasing pH is thought to be result of the liberation of OH^- in the lysine-buffered sol during germania condensation that accompanies crystallization in a way similar to zeolite crystal growth.[24] Bragg reflections measured by SWAXS (Figure 5a)

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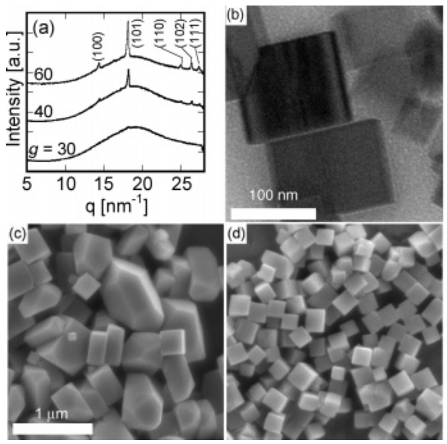


Figure 5. (a) SWAXS data for three sols of region II′ (Figure 3a) of molar composition g GeO_2/5.8 lysine/9500 H_2O/4g ethanol, where $g = 30$, 40, and 60. Reflections corresponding to the crystalline hexagonal phase of germania are labeled. In situ cryo-TEM (b) and ex situ SEM (c) images of germania crystals formed in a Lys-Ge sol of molar germania content $g = 30$. Panel (d) shows an SEM image of a crystalline precipitate formed in a Lys-Ge sol of molar composition $g = 40$. Images in (c) and (d) are at the same magnification.

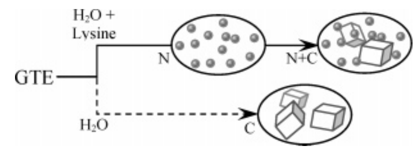
at large q values for these sols share identical positions to the low-angle peaks of the complete diffraction patterns obtained via powder XRD of the dried Lys-Ge sols (Figure S2, Supporting Information). This confirms the presence of crystalline, hexagonal-phase germania particles coexisting in the sols with the 1 nm particles at room temperature.

Representative SEM images of dried precipitates from several Lys-Ge sols are shown in Figure 5c,d. The crystals are noticeably smaller, more cubic, and apparently more uniform in size and shape when formed at $g = 40$ (Figure 5d) as compared to $g = 30$ (Figure 5c), for which large particles (~200−1000 nm) display a polyhedral morphology. The particle size continues to decrease with further increases in germania concentration (Figure S3, Supporting Information). At all compositions studied, the dried crystals appear to be discrete particles and are characterized by well-defined crystal faces with smooth surfaces.

Similar to the lysine-free water−germania sols, a rise in the low-q scattering occurs for Lys-Ge sols of lower germania concentration in region II′ (e.g., $g = 25$, 30; Figure 3b), but Bragg reflections are not observed with SWAXS (e.g., $g = 30$; Figure 5a). SEM images of the dried sols (Figure 5c,d), however, depict crystals that yield a powder XRD pattern consistent with the hexagonal phase of germania. Cryo-TEM (Figure 5b) of the same sols reveals crystals in solution that are of a comparable distributed size and morphology to those in the dried state (i.e., compare with SEM of Figure 5c). Therefore, the absence of Bragg reflections from SWAXS for these sols seems to be an artifact of the concomitantly small crystal concentration, ruling out the possibility of concentration-induced crystal formation upon evaporative drying for SEM imaging.

Ultimately, the presence of lysine during hydrolysis of GTE perturbs the regime boundaries established in the lysine-free water−germania system (Figure 1a) in a number of ways. The presence of lysine leads to the introduction of a nanoparticle phase in region I and modification of region II to include not only crystals, but also nanoparticles. The differences between crystallization in Lys-Ge sols and germania in pure water are summarized in Scheme 1. Lysine appears to delay the onset of crystal formation. Namely, in the presence of lysine, crystals form only at germania compositions of more than twice that in pure water (i.e., 0.14 M vs 0.06 M, respectively). A mechanism cannot be conclusively defined here for germania crystallization

Scheme 1. Routes to Germania Crystallization Involving Nanoparticles (N) and Crystals (C) with Germania Content Increasing Left to Right



either in the presence or absence of lysine. We speculate, however, that lysine may stabilize germanate species through complexation observed via ESI-MS, giving rise to stable and detectable germania nanoparticles. Such stabilization may contribute to the delay in crystal nucleation and growth, requiring additional (unstable) germania species (i.e., higher germania concentrations) before crystal nucleation can be initiated.

The germania crystals in the lysine-free water−germania and Lys-Ge system, and the nanoparticles in sols of the latter system, appear to be inherently stable to aggregation over the time scales investigated here. While investigation of the particle stability for germania nanoparticles and crystals is beyond the scope of this work, a number of factors may lead to particle stability in this system, including (1) the potential association of lysine with the surface of the nanoparticles as described in ref 25, (2) the inert character of the nanoparticles enabling redispersion of formed aggregates even by weak Brownian forces, and/or (3) the particle morphology (i.e., possibly nonspherical), hindering close contact required for aggregation.

Ultimately, the germania crystal morphology observed by TEM and SEM shows no evidence of nanoparticle aggregation (e.g., particulate nature, crystal faults, misoriented crystal domains) as a means for crystal formation. Instead, germania crystallization seems to be dominated by rapid nucleation and growth without the direct participation of nanoparticles over the time scales investigated here. As such, germania crystallization appears to be remarkably different than the aggregative mechanisms common to other oxide systems. Namely, crystallization has been observed under hydrothermal conditions to occur over hours by oriented aggregation of crystalline nanoparticles (e.g., TiO_2[26] and Fe_2O_3[27]) or by aggregation of nanoparticles (e.g., silicalite-1[7]) that have evolved in their colloidal and structural stability toward a capacity for growth over time scales of months at room temperature. It is conceivable that the rapid crystallization in the germania system precludes participation by the germania nanoparticles especially over short times. The relatively short-term studies performed here, however, cannot discount the possibility of secondary crystallization by aggregation of aged germania nanoparticles.

This work represents, to our knowledge, the first identification of an oxide system that undergoes rapid room-temperature crystallization in both the presence and absence of nanoparticles. The room temperature and near-neutral conditions for crystallization, especially in the presence of the basic amino acid lysine, make this rapid oxide crystallization attractive for benign materials applications.

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Supporting Information Available: Supplemental SEM images of crystalline precipitates and associated powder XRD patterns. The material is available free of charge via the Internet at http://pubs.acs.org.

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